

Semantic Playlist Representation and Recommendation

Using BERT Embeddings, PCA, UMAP, and HDBSCAN

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Abstract—This paper investigates whether short Spotify playlist titles contain enough semantic information to support visualization, clustering, and cold-start retrieval. Using the Spotify Million Playlist Dataset (MPD), we represent each playlist with two paired 384-dimensional Sentence-BERT embeddings: one for the playlist title and one mean-pooled embedding derived from its track titles [1], [2]. This paired design lets us compare the user’s own description of the playlist with the playlist’s observed content while assigning different roles to the two representations: mean-pooled track embeddings serve as the primary space for dimensionality reduction and clustering, while title embeddings provide validation and retrieval signals. We analyze the track-embedding space using PCA, t-SNE, UMAP, K-means, and HDBSCAN, and we compare cluster-first workflows in higher-dimensional reduced spaces with reduce-first workflows that cluster directly on two-dimensional coordinates. Clustering in very low-dimensional spaces produced visually cleaner groups and higher apparent cluster quality, but these solutions discarded substantial variance and likely introduced artificial cluster structure. By contrast, intermediate-dimensional representations preserved more of the original embedding information while still allowing clustering to reveal interpretable patterns, making them a more defensible basis for analysis. We also find that cosine similarity on the embeddings supports plausible cold-start retrieval, but this recommendation signal is largely independent of dimensionality reduction and clustering. Overall, the results suggest that playlist titles contain useful but limited semantic information: enough to support retrieval and exploratory visualization, but not enough for low-dimensional clustering results to be freely interpreted as genuine structure.

I. INTRODUCTION

Modern music streaming platforms serve hundreds of millions of users with highly individualized listening preferences. Conventional recommendation systems address this complexity by using behavioral signals, such as listening history, skips, saves, and ratings, together with item-level features such as genre labels, artist metadata, and audio descriptors [3]. These systems work well when a platform has sufficient user or playlist information, but they face a cold-start problem when a user creates a new playlist, forcing the system to infer preferences from limited available context.

Playlist titles provide one such source of context because prior playlist-generation work has shown that title text can provide useful semantic constraints for recommendation [4]. For example, when a user names a playlist “Late Night Chill Vibes” or “Morning Workout Bangers”, the title can encode genre, mood, activity, energy level, or intended listening environment in a short textual phrase. This project asks whether playlist titles contain enough semantic signal to support cold-start retrieval and interpretable visualization. In particular, we

study whether playlist titles align with the actual track content of the playlists they describe.

This question has both a recommendation component and a geometric visualization component. On the recommendation side, we ask whether short text queries such as “workout,” “break up,” or “rainy day” can retrieve related songs or playlists. On the visualization side, we ask whether high-dimensional text embeddings reveal interpretable structure after dimension reduction. A useful cold-start representation should not only retrieve plausible recommendations, but also organize playlists in a way that reflects semantic or musical similarity.

To investigate this problem, we use Sentence-BERT (SBERT) to construct two aligned 384-dimensional representations for each playlist: one encoding the playlist title and one mean-pooled representation of its track titles [5]. This paired design lets us compare the user’s short textual description with the playlist’s actual track content, using mean-pooled track embeddings as the primary space for dimensionality reduction and clustering and title embeddings as validation and retrieval signals.

We analyze the mean-pooled track embedding space using Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), Uniform Manifold Approximation and Projection (UMAP), K-means, and HDBSCAN [6]–[8]. PCA provides a global linear baseline, t-SNE and UMAP provide nonlinear views of local neighborhood structure, and the clustering methods allow us to compare centroid-based and density-based grouping behavior. We then evaluate whether the resulting track-based structure aligns with playlist intent by comparing matched title-track embeddings with random mismatched pairs and by measuring title coherence within track-based clusters.

A central methodological question in this report is whether visually clean low-dimensional structure reflects the higher-dimensional embedding geometry or instead introduces spurious information that confounds the clustering process. To address this, we compare cluster-first workflows, in which clustering is performed in a higher-dimensional reduced space before visualization, with reduce-first workflows, in which clustering is performed directly on two-dimensional PCA or t-SNE coordinates.

Overall, this report argues that playlist titles contain useful but limited semantic information. Cosine-similarity retrieval provides a plausible cold-start signal, whereas low-dimensional clustering can produce visually appealing but potentially misleading structure. Section II reviews the relevant

background, Section III describes the dataset and preprocessing steps, Section IV presents the methodology, Section V reports the results, Section VI discusses the findings and limitations, and future research, and Section VII concludes.

II. BACKGROUND AND RELATED WORK

A. The Cold-Start Problem in Recommendation

Collaborative filtering systems build user-item interaction matrices and identify similar users or items based on shared behavioral patterns. While powerful, these methods require substantial interaction data. The cold-start problem refers to the degraded performance that occurs when this data is absent [3]. Content-based filtering partially addresses the issue by relying on item attributes rather than user behavior, but it typically requires structured metadata such as genre tags or audio features that may not always be available or reliable. Even when such information is available, it may be difficult to convert into a form that is useful for recommendation.

Title-based semantic retrieval occupies a middle ground: it requires no behavioral data at query time and uses the natural language content of the playlist name as a lightweight cold-start signal. Prior work by Charolois-Pasqua et al. showed that language models can generate reasonable playlist continuations using only title prompts, suggesting that playlist titles encode useful semantic constraints on song selection [4].

B. SBERT and Sentence Embeddings

BERT (Bidirectional Encoder Representations from Transformers) is a pretrained language model that produces contextualized token-level embeddings by attending to the full left and right context of each token simultaneously. For sentence-level tasks, a common approach is to mean-pool the token embeddings, producing a single vector that summarizes the entire input. SBERT extends BERT by using Siamese and triplet network structures to produce sentence embeddings that can be compared efficiently with cosine similarity [5].

This approach captures not just lexical overlap between titles but their semantic intent. The title embedding encodes the user-written playlist name, while the mean-pooled track embedding averages the SBERT embeddings of the playlist's track titles. We use the track-based representation for clustering and dimension reduction, and we use the title representation to evaluate whether track-based clusters align with playlist intent. This paired representation lets us compare two related but distinct signals: the user's stated intent in the playlist title and the observed content summarized by the playlist's track titles.

C. Dimensionality Reduction

The 384-dimensional embedding space is high-dimensional and difficult to visualize or analyze directly. Dimensionality reduction methods project data into lower-dimensional spaces while attempting to preserve meaningful structure at different scales. We employ three dimension reduction techniques here.

- PCA is a linear method that finds orthogonal directions of maximum variance and projects the data onto the top

k principal components [6]. PCA is computationally efficient, globally interpretable, and provides an information-theoretic baseline through the fraction of variance explained by the retained components.

- t-SNE is a nonlinear method designed to preserve local neighborhood relationships in low-dimensional visualizations of high-dimensional data [9]. Unlike PCA, its axes do not have direct semantic interpretation; the meaningful information lies mainly in which points appear near one another. We therefore treat t-SNE primarily as a visualization tool rather than as a clustering ground truth. To assess how well a t-SNE map preserves local neighborhoods from the higher-dimensional space, we use trustworthiness.
- UMAP is another nonlinear dimensionality reduction method based on ideas from topology and manifold learning. Compared with PCA, UMAP often emphasizes local neighborhood structure more strongly and can produce visually clearer cluster-like regions, but it also sacrifices direct axis interpretation and can distort global distances. Because low-dimensional projections can make structure appear cleaner than it is in the original embedding space, we treat these methods not only as visualization tools but also as objects of methodological comparison later in the report [7].

D. Clustering Methods

K-means serves as a centroid-based baseline. Given a fixed number of clusters k , it partitions points by minimizing within-cluster squared distance to cluster centroids. This makes K-means computationally efficient and easy to compare across embedding spaces, but it also assumes roughly convex clusters and requires the number of clusters in advance. In this project, we use K-means mainly as a reference method for comparing cluster-first and reduce-first workflows.

We also use HDBSCAN (Hierarchical DBSCAN) as a density-based clustering method. HDBSCAN extends DBSCAN by building a hierarchy of clusters across density thresholds and extracting stable clusters from that hierarchy [8]. Unlike K-means, HDBSCAN does not require specifying the number of clusters in advance, can accommodate clusters of irregular shape, and explicitly labels low-density points as noise (-1). These properties make it well-suited for exploratory clustering in nonlinear embeddings, where playlist groups may have uneven densities and non-convex structure.

III. DATASET AND PREPROCESSING

A. Spotify Million Playlist Dataset

The Spotify Million Playlist Dataset (MPD), released through the AICrowd RecSys Challenge 2018, contains one million user-created playlists collected from the Spotify platform between January 2010 and October 2017 [1], [2]. Each entry in the dataset includes a human-assigned playlist title, an ordered list of track names, and associated metadata such as artist names and album titles. It was designed as a benchmark for automatic playlist continuation tasks and has since become a standard resource in music information retrieval research [2].

For the final embedding analysis, we used a cleaned embedding file organized as a row-aligned table. Each row represents one playlist and contains four main fields: `playlist_title`, `title_embedding`, `playlist_tracks`, and `tracks_embedding`. The `title_embedding` column stores a 384-dimensional SBERT representation of the playlist title, while the `tracks_embedding` column stores a 384-dimensional mean-pooled representation constructed from the playlist’s track titles. This structure lets us compare each playlist’s user-written title with a track-derived representation of its actual content.

Because several visualization and clustering methods scale poorly to one million playlists, we used a fixed random sample of 20,000 playlists for PCA, t-SNE, K-Means, workflow comparisons, and validation metrics. We used the same sampled rows for title embeddings and mean-pooled track embeddings, so the two matrices remained aligned playlist by playlist. After preprocessing, both sampled matrices had shape $20,000 \times 384$, giving each playlist one 384-dimensional title vector and one 384-dimensional track-content vector. Unless otherwise noted, exploratory visualizations, workflow comparisons, and validation analyses use this fixed 20,000-playlist sample, while the larger UMAP-HDBSCAN pipeline uses the higher-dimensional fitting and transformation steps described in Section IV.

B. Exploratory Statistics

The full dataset contains 1,000,000 playlists. Several summary statistics motivate the paired embedding design used in the later analysis. Playlist titles are short: the mean title length is 7.7 characters, the median is 7.0 characters, and the range is 1 to 100 characters. This indicates that most users write short functional labels rather than descriptive sentences. By contrast, playlist size varies much more substantially. The mean playlist contains 66.3 tracks, the median contains 49.0 tracks, and the range is 5 to 376 tracks. The difference between the mean and median indicates a right-skewed distribution, where a relatively small number of long playlists pull the average upward.

TABLE I
DATASET SUMMARY

Dataset property	Value / description
Number of playlists	1,000,000
Approximate number of tracks	66 million
Approximate number of unique tracks	2 million
Approximate number of artists	295,000
Approximate number of albums	734,000
Mean playlist title length	7.7 characters
Median playlist title length	7.0 characters
Minimum playlist title length	1 character
Maximum playlist title length	100 characters
Mean tracks per playlist	66.3 tracks
Median tracks per playlist	49.0 tracks
Minimum tracks per playlist	5 tracks
Maximum tracks per playlist	376 tracks
Embedding dimension	384
Sample size used for embedding experiments	20,000 playlists

Table I summarizes the main dataset properties and the sample used for the embedding experiments. Playlist titles are short and sparse compared with playlist contents. This contrast motivates the paired design: title embeddings represent the cold-start query signal, while mean-pooled track embeddings represent richer playlist content.

These distributions are shown in Figure 1. Playlist titles are concentrated around short, often single-word labels, whereas playlist track counts are strongly right-skewed. In cumulative terms, 50% of playlists contain fewer than 50 tracks, 78% contain fewer than 100 tracks, and 97% contain fewer than 200 tracks. Together, these patterns motivate the paired representation used throughout the report: titles provide a sparse cold-start text signal, while playlists themselves provide much richer content through their track lists.

The most common lowercased playlist titles are short genre-, mood-, or activity-oriented labels, including “chill” (9,256 playlists), “country” (8,724), “rap” (8,045), “workout” (7,644), and “oldies” (7,413). Figure 2 shows that these frequent titles cluster around a small number of broad semantic categories. This supports the intuition that playlist titles contain meaningful signal, but it also shows their sparsity: in many cases, a single word must represent the intended genre, mood, or activity.

C. Preprocessing

We turned all playlist titles to lowercase to merge case variants (e.g., “Chill” and “chill” are treated as identical). We did not apply additional stop-word removal or stemming, since SBERT’s subword tokenization handles morphological variation internally. We constructed two embedding representations for each playlist. First, we encoded the playlist title with SBERT to obtain a title embedding. Second, we encoded the playlist’s track titles and averaged those track-title embeddings to obtain a mean-pooled track-based playlist embedding. The title embedding represents the user’s short textual description, while the mean-pooled track embedding represents the playlist’s observed song content. This paired representation allows us to test whether title semantics align with the track-derived content of the playlists they describe. We normalized embeddings only for cosine-similarity comparisons. We kept the non-normalized mean-pooled track embeddings for the main UMAP and HDBSCAN clustering pipeline so that the track-content geometry retained useful magnitude variation. This design separates the cold-start signal from the content signal: the title embedding represents what the user wrote, while the mean-pooled track embedding represents what the playlist contains. Comparing the two reveals how much information the title carries about the playlist’s musical content.

We do not treat the two embedding types as interchangeable clustering inputs. A diagnostic norm check showed that title embeddings have constant unit norm, while mean-pooled track embeddings have variable norms. This difference changes the behavior of Euclidean distances, PCA, UMAP, and density-based clustering. For that reason, we cluster mean-pooled track embeddings and use titles as validation signals.

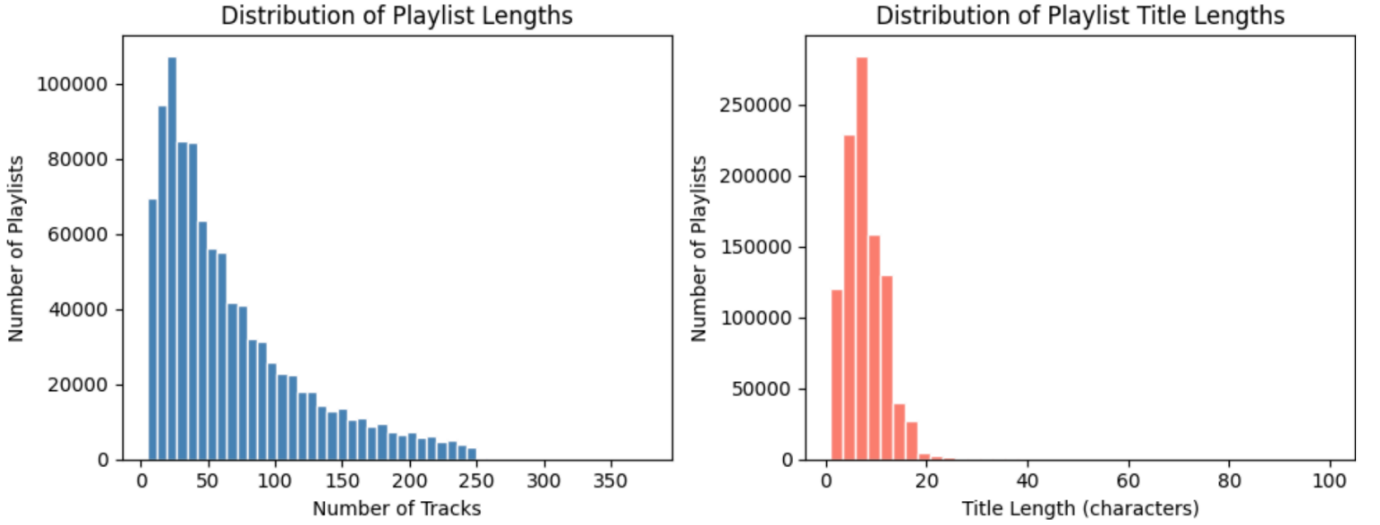


Fig. 1. Distribution of playlist track counts and title lengths counts. Playlist titles tend to have short, often single-word titles, whereas playlist track counts are strongly right-skewed. Half of playlists contain fewer than 50 tracks, 78% contain fewer than 100 tracks, and 97% contain fewer than 200 tracks.

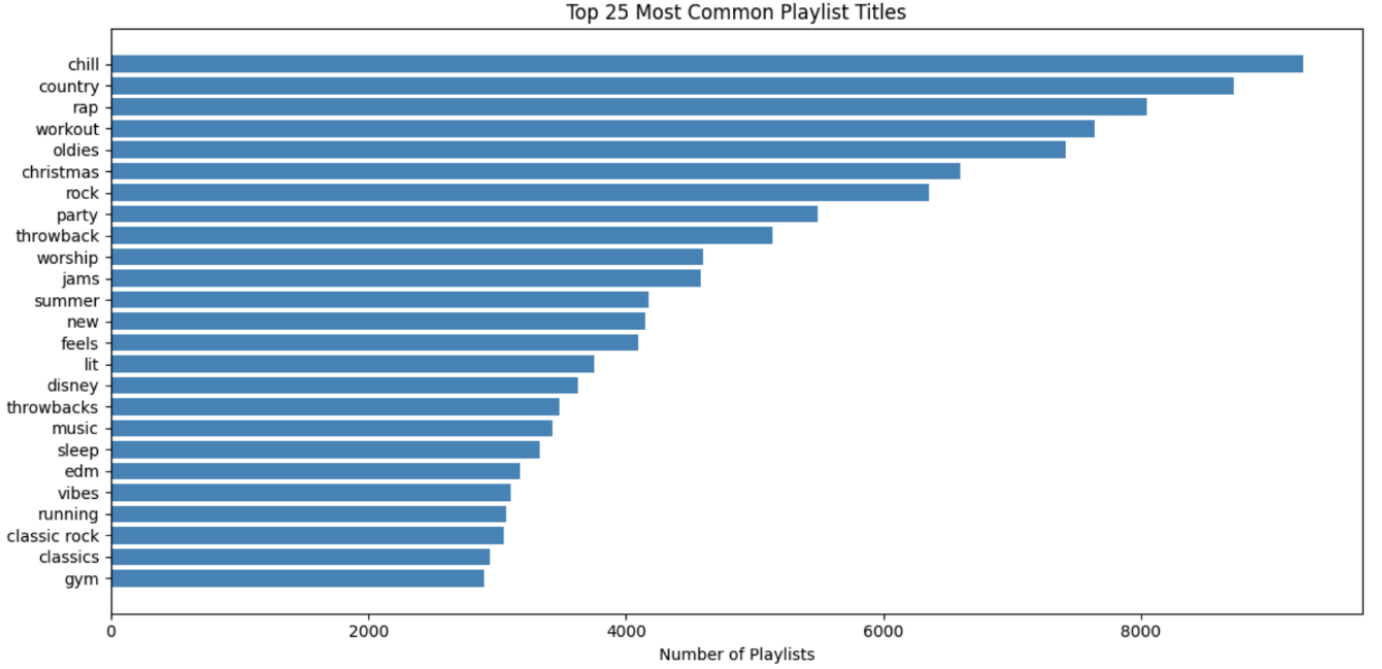


Fig. 2. Top 25 most common lowercased playlist titles in the dataset. The most frequent titles are short genre-, mood-, and activity-based labels, showing both the semantic relevance and the sparsity of playlist-title text.

IV. METHODS

A. Embedding Representation

We represent each playlist with two aligned 384-dimensional SBERT embeddings: one for the title and one mean-pooled embedding for the track titles [5]. We use the mean-pooled track embedding as the primary object for dimension reduction and clustering. This approach, following Reimers and Gurevych, places playlists in a semantic text-embedding space where distance reflects similarity in the language used to describe playlist titles and track titles. Because the mean-pooled track embedding averages the track-title representations in a playlist, nearby mean-pooled track

embeddings suggest similar textual content across playlists, not necessarily direct acoustic similarity [5].

The mean-pooling operation aggregates information across all tracks in a playlist, producing a representation that reflects the average semantic profile of the playlist’s track-title text. Formally, if playlist p contains n_p tracks with embeddings $t_1, \dots, t_{n_p} \in \mathbb{R}^{384}$, then its mean-pooled track embedding is

$$\bar{t}_p = \frac{1}{n_p} \sum_{j=1}^{n_p} t_j.$$

This vector represents the centroid of the playlist’s track-title embeddings.

We normalized embeddings for cosine-similarity comparisons, but we retained the non-normalized mean-pooled track embeddings for the main clustering pipeline because their norm variation contributed to the observed track-content geometry [5].

We chose not to cluster title embeddings in the final pipeline. The title embeddings had nearly identical L2 norms, which placed them on a unit sphere and made PCA and Euclidean-distance-based clustering less informative. Mean-pooled track embeddings, by contrast, retained more geometric variation after mean pooling and therefore provided a more useful clustering signal.

B. Title-Content Alignment Metrics

To evaluate whether playlist titles contain information about playlist content, we compared each title embedding with the mean-pooled track embedding from the same playlist. We then compared these matched title-track cosine similarities against random mismatched pairs, providing a direct quantitative test of whether a playlist title lies closer to its own track-derived content representation than to unrelated content.

After clustering the mean-pooled track embeddings, we measured title coherence within each cluster using pairwise cosine similarity ($\cos(\theta) = \frac{\mathbf{v}_q \cdot \mathbf{v}_i}{\|\mathbf{v}_q\| \|\mathbf{v}_i\|}$) among title embeddings. This does not provide ground-truth genre labels, but it tests whether playlists grouped by track content also have semantically similar names.

C. Principal Component Analysis

We applied PCA to the track-embedding matrix to study the dominant linear directions of variation. For computationally intensive comparisons, we used fixed samples; for the UMAP-HDBSCAN pipeline, we fit UMAP on a 100,000-playlist sample and transformed the full one-million-playlist dataset in batches. PCA decomposes the centered data matrix X as $X = U\Sigma V^T$, where the columns of V are the principal components and the diagonal entries of Σ are the corresponding singular values [6].

We used PCA in two ways. First, PCA-50 served as a higher-dimensional reference space for K-Means workflow comparisons, because it reduced the original 384-dimensional embeddings while preserving about 65% of the track-embedding variance. Second, PCA-2 and PCA-3 provided linear visualization baselines. We did not treat PCA visual clusters as ground-truth categories; instead, we used PCA to study global variance structure and to compare how clustering behavior changed after aggressive two-dimensional reduction. The explained-variance analysis showed that the first few components capture important variation, but the track-embedding space remains genuinely high-dimensional, with substantial structure distributed across many dimensions.

PCA visualization revealed broad structure and interpretable extremes but did not provide definitive cluster labels. Consequently, we treated PCA primarily as a linear baseline and diagnostic tool rather than as evidence for ground-truth genre clusters.

D. t-SNE and K-Means Workflow Comparison

We used t-SNE as a nonlinear visualization of local neighborhood structure after reducing the mean-pooled track embeddings to PCA-50 [9]. Because t-SNE axes do not have direct semantic meaning, we interpreted t-SNE plots through local neighborhoods rather than through coordinate values. We also tested perplexity sensitivity and computed trustworthiness to evaluate how well the two-dimensional t-SNE maps preserved local neighborhoods from PCA-50 space.

We used K-means as a reference method for comparing clustering workflows. In the cluster-first workflow, we fit K-means with $k = 50$ in PCA-50 space, then visualized the resulting labels in PCA-2 or t-SNE-2. In the reduce-first workflow, we first reduced the data to PCA-2 or t-SNE-2 and then fit K-means directly on the two-dimensional coordinates. We evaluated agreement between these partitions primarily using the Adjusted Rand Index (ARI), in order to test whether visually clean two-dimensional clusters preserved the same grouping structure as clustering in a higher-dimensional representation [?].

E. UMAP

We applied UMAP to mean-pooled track embeddings as the main nonlinear reduction method [7]. We tested 10-, 15-, and 20-dimensional UMAP outputs as candidate clustering spaces and used a separate two-dimensional UMAP projection only for visualization. We selected the 15-dimensional UMAP representation as the main HDBSCAN clustering space because, in diagnostic runs, it produced a useful balance between preserving local structure and yielding interpretable density-based clustering behavior. For the visualization projection, we used UMAP with 15 neighbors and minimum distance of 0.1. The two-dimensional UMAP plots therefore display the cluster labels, but they do not define the clusters.

UMAP often emphasizes local manifold structure more strongly than PCA in two-dimensional projections, which can make cluster-like regions appear more visually separated. PCA preprocessing can improve nonlinear manifold learning in some settings, but in the final HDBSCAN pipeline we distinguished between the representation used for clustering and the two-dimensional projection used for display.

The UMAP embedding of the track-based playlist space showed visible density structure and separated regions, but we interpret those regions through title-based validation rather than assuming they correspond directly to formal genre or mood labels. Manual inspection suggested that nearby UMAP regions often contained playlists with related titles or track-title content, but we treated this as qualitative evidence rather than formal genre validation.

F. HDBSCAN Clustering

We applied HDBSCAN to the 15-dimensional UMAP mean-pooled track embedding [8]. We used a 2D UMAP projection to display the resulting labels. The key hyperparameters are `min_cluster_size` (the minimum number of points that constitutes a cluster) and `cluster_selection_epsilon` (which merges clusters

that are too close to each other). We explored a range of values for both parameters, evaluating results both quantitatively (via the relative validity metric) and qualitatively (via inspection of cluster titles and tracks).

HDBSCAN assigns a label of -1 to noise points—playlists that do not belong to any sufficiently dense cluster. We interpret noise points as playlists that HDBSCAN does not place into a sufficiently dense region of the track-embedding space. Some may represent ambiguous, mixed-theme, or idiosyncratic playlists, but the noise label reflects density under the chosen representation rather than a confirmed semantic category. Tracking the noise fraction provides a useful diagnostic for the quality of the embedding and UMAP projection.

G. Nearest-Neighbor Retrieval

Operationally, the retrieval procedure consisted of four steps. We first standardized and embedded the query playlist title using SBERT. We then computed cosine similarity between the query embedding and the candidate embedding set, retrieved the top-K nearest playlists, and sampled songs from those retrieved playlists to form candidate recommendations. We treat this as a qualitative cold-start retrieval workflow rather than a complete deployed recommendation system [9].

For the retrieval task, we implemented cosine similarity search using normalized embeddings. Given a query playlist title, we encoded the title with SBERT and compared it against candidate playlist or track-derived embeddings using cosine similarity. Cosine similarity compares vector direction rather than magnitude, making it appropriate for high-dimensional text embeddings. We interpret the returned neighbors as semantically related candidates rather than as a complete recommendation system.

Operationally, the retrieval workflow proceeded as follows: a user-provided playlist title was first standardized and embedded with SBERT, cosine similarity was then computed between that query embedding and the candidate embedding set, the top-K nearest playlists were retrieved, and songs were sampled from those retrieved playlists to produce candidate recommendations. This workflow was used as a qualitative cold-start retrieval procedure rather than as a full recommendation pipeline.

For computational tractability, we implemented the cosine search over a 5,000-playlist sample using matrix multiplication after normalizing all embeddings to unit length, reducing the pairwise similarity computation to a single matrix multiply. This yields exact nearest neighbors within the sampled candidate set and makes qualitative retrieval experiments computationally feasible.

V. RESULTS

A. Embedding Geometry Diagnostic

Before clustering, we compared the norm structure of title and mean-pooled track embeddings. Title embeddings had mean L2 norm 1.000 with standard deviation 0.000, while mean-pooled track embeddings had mean norm 0.459 with standard deviation 0.041. This diagnostic motivated the final design choice: we used mean-pooled track embeddings for

dimension reduction and clustering, and title embeddings for validation and retrieval. This avoids forcing the two representations into a single clustering geometry when their distance behavior differs substantially, and it motivates the use of title embeddings primarily as external validation signals.

We also evaluated how closely playlist titles aligned with playlist content by comparing each title embedding with its own mean-pooled track embedding and with randomly mismatched mean-pooled track embeddings from other playlists. Matched title-track pairs were modestly but consistently more similar than random mismatched pairs. The mean cosine similarity was 0.436 for matched pairs and 0.399 for random mismatches, a difference of 0.037, and the probability that a matched pair exceeded a random pair was 65.4%. This result indicates that playlist titles contain measurable semantic information about playlist content, but the effect is not strong enough to treat title text alone as a complete representation of playlist structure. Accordingly, title embeddings were used primarily as validation and retrieval signals rather than as the main clustering space.

B. Low-Dimensional Views of the Track Embedding Space

Before comparing clustering workflows, we examined how the mean-pooled track embeddings appear under three common two-dimensional projections: PCA, t-SNE, and UMAP. PCA provided a coarse global linear summary with one dense central region and a smaller number of playlists spread toward the extremes. By contrast, t-SNE and UMAP emphasized local neighborhood structure and produced more visibly separated regions. These projections are useful for visualization, but they do not define equivalent clustering spaces; later results show that cleaner two-dimensional separation does not necessarily preserve the higher-dimensional partition.

PCA helped identify dominant directions of variation in the track-embedding space and provided interpretable examples near the extremes of the principal axes. However, PCA did not serve as the final clustering space in the UMAP-HDBSCAN pipeline. We used it as a linear baseline and as a diagnostic for whether mean-pooled track embeddings contained visible large-scale structure. The PCA extremes were interpretable, but they did not define clean clusters. Playlists on the positive side of PCA-1 tended to have generic titles such as fave or symbol-only names, whereas the negative side of PCA-1 contained more specific labels such as rancheras, salsa, old fashion, and calibre 50. The upper and lower extremes of PCA-2 also showed thematic differences, with titles such as old fashion and calibre 50 at the top and electro and christmas at the bottom. These patterns suggest that the leading principal components capture broad differences in title specificity and theme, but they remain a coarse linear summary rather than definitive category boundaries.

Because cumulative explained variance increases gradually rather than reaching an early plateau, the mean-pooled track embedding space appears to remain genuinely high-dimensional. Although very low-dimensional PCA reductions produce higher apparent silhouette, they also discard much of the original embedding information. We therefore use

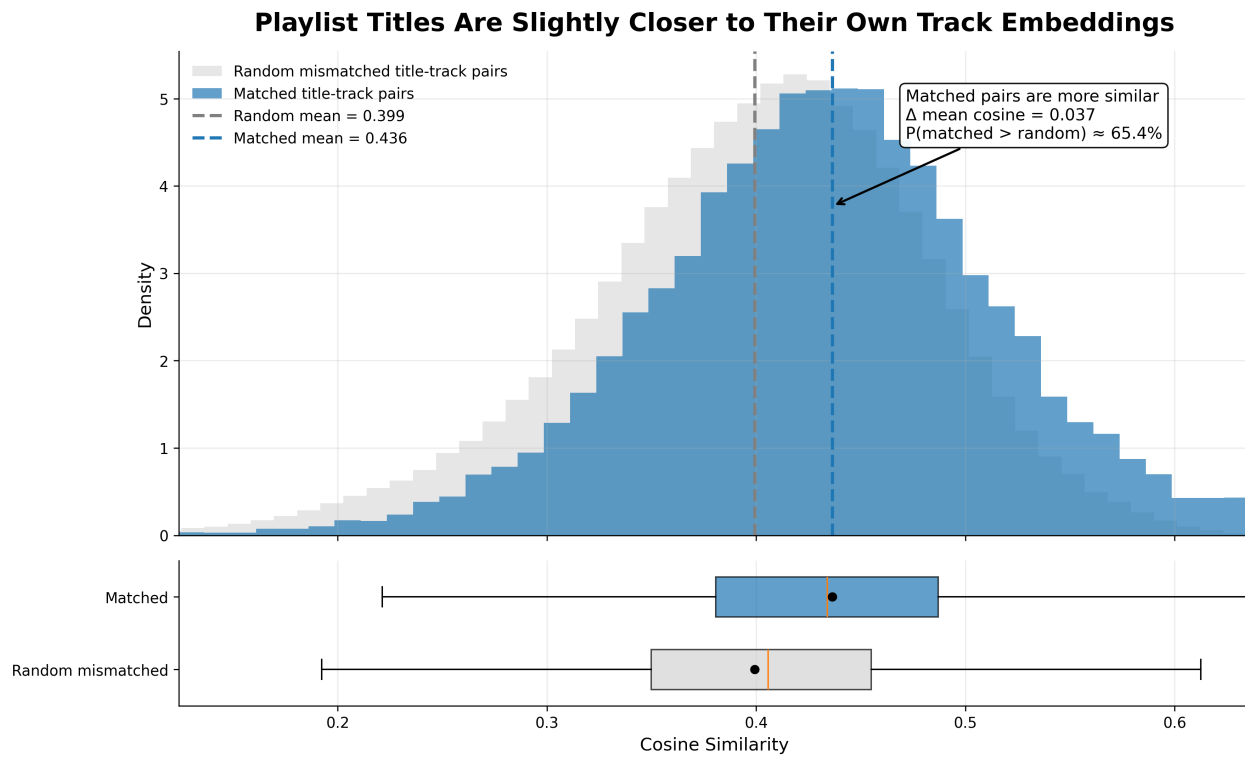


Fig. 3. Cosine similarity for matched and random mismatched title-track pairs. Matched pairs are modestly more similar on average, indicating that playlist titles contain measurable but limited information about playlist content. The substantial overlap between distributions shows that title embeddings alone do not strongly explain the mean-pooled track embedding structure.

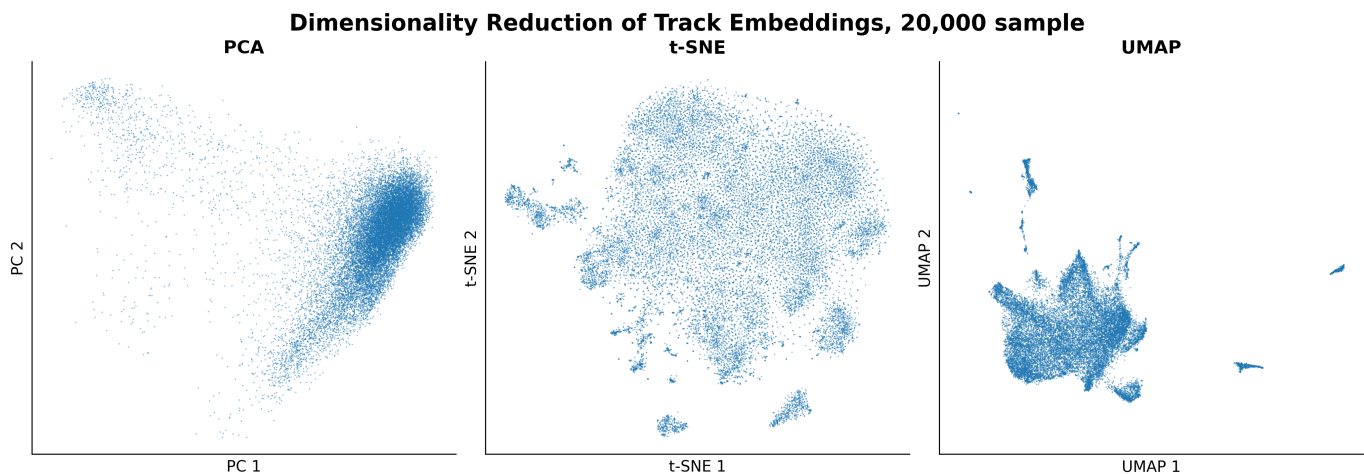


Fig. 4. Two-dimensional views of the mean-pooled track embeddings under PCA, t-SNE, and UMAP. PCA provides a global linear summary of the embedding space, while t-SNE and UMAP emphasize local neighborhood structure and produce increasingly separated visual regions. These plots are useful for visualization, but the apparent 2D separation should not be interpreted as definitive cluster structure without further comparison.

Track PCA Extremes: Example Playlists Along the Main Axes

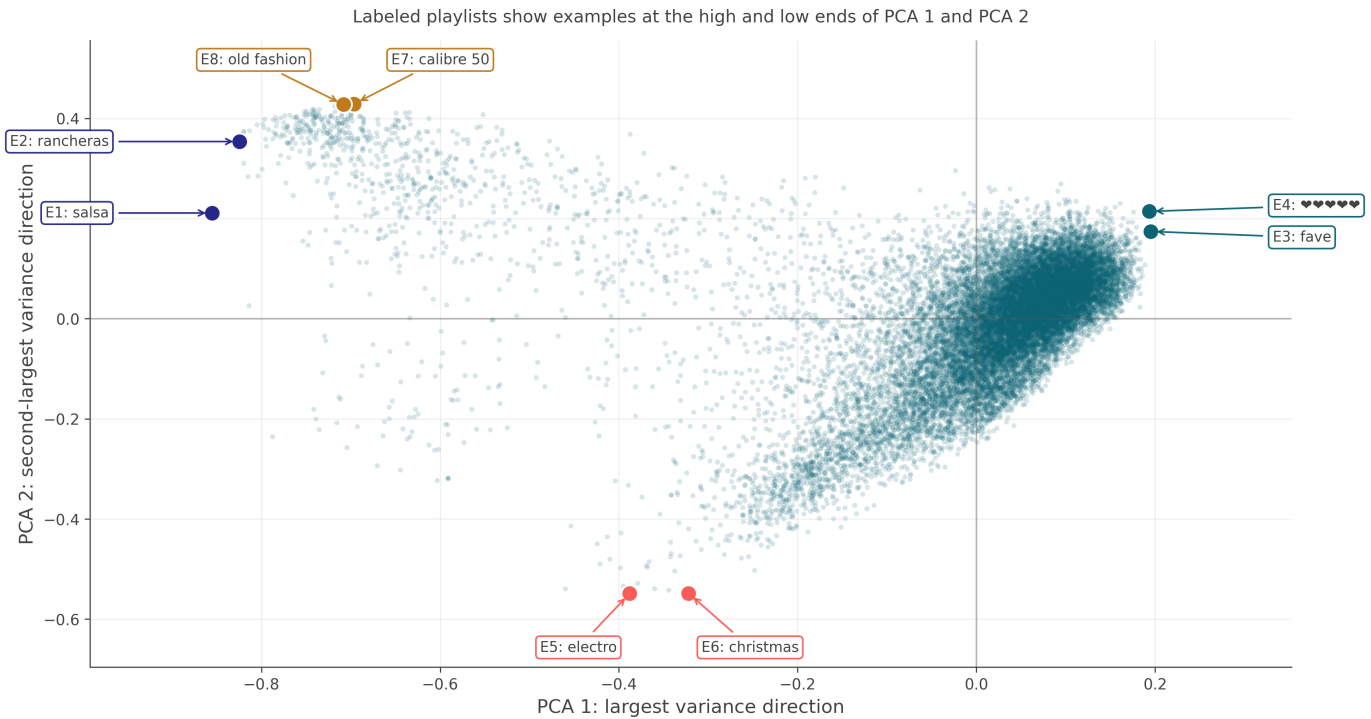


Fig. 5. PCA projection of mean-pooled track embeddings with labeled playlists from the extremes of PCA-1 and PCA-2. While the dense center reflects broad overlap, the extremes spotlight more distinctive title themes. These labels provide qualitative clues about the main PCA directions, not exact definitions of the axes.

PCA-50 as a compromise between information retention and dimensionality reduction before comparing cluster-first and reduce-first workflows.

C. Clustering on reduced dimensions

Because cumulative explained variance increases gradually rather than reaching an early plateau, the mean-pooled track embedding space appears to remain genuinely high-dimensional. Although very low-dimensional PCA reductions produce higher apparent silhouette, they also discard much of the original embedding information. We therefore use PCA-50 as a compromise between information retention and dimensionality reduction before comparing cluster-first and reduce-first workflows.

We then compared cluster-first and reduce-first workflows using PCA-50 as the higher-dimensional reference space. This comparison showed that clustering directly on two-dimensional embeddings produced visually cleaner groups, but substantially altered the underlying partition relative to the higher-dimensional reference.

We then compared cluster-first and reduce-first workflows using PCA-50 as the higher-dimensional reference space. On the PCA maps, the cluster-first workflow–K-means fit in PCA-50 and displayed in PCA-2–produced broad overlap among labels, especially in the dense central wedge of the embedding. By contrast, the reduce-first workflow–K-means fit directly on PCA-2–produced compact and visually distinct regions.

However, these cleaner PCA-2 clusters were not faithful to the higher-dimensional reference partition: the adjusted Rand index (ARI) between K-means on PCA-2 and K-means on PCA-50 was only 0.103.

The t-SNE comparison led to the same qualitative conclusion. When PCA-50 cluster labels were displayed on the two-dimensional t-SNE map, the labeled regions remained interwoven and locally mixed. When K-means was fit directly on the two-dimensional t-SNE coordinates, the result was a set of visually distinct, blob-like clusters. Even so, agreement with the PCA-50 reference remained low, with an ARI of 0.250. Thus, clustering directly in two dimensions changed the partition substantially even when the plots appeared cleaner.

Taken together, these results show that low-dimensional clustering can produce higher apparent cluster quality while simultaneously discarding substantial information and inducing artificial separation. For that reason, we used higher-dimensional reduced spaces for clustering and reserved two-dimensional projections for visualization and interpretation.

D. HDBSCAN Cluster Quality

The two-dimensional UMAP projection provided the clearest visual summary of local neighborhood structure in the mean-pooled track embeddings, so we used UMAP as the basis for the final density-based clustering stage. The 2D map showed visibly separated regions associated with interpretable title themes, including worship and christian playlists, workout

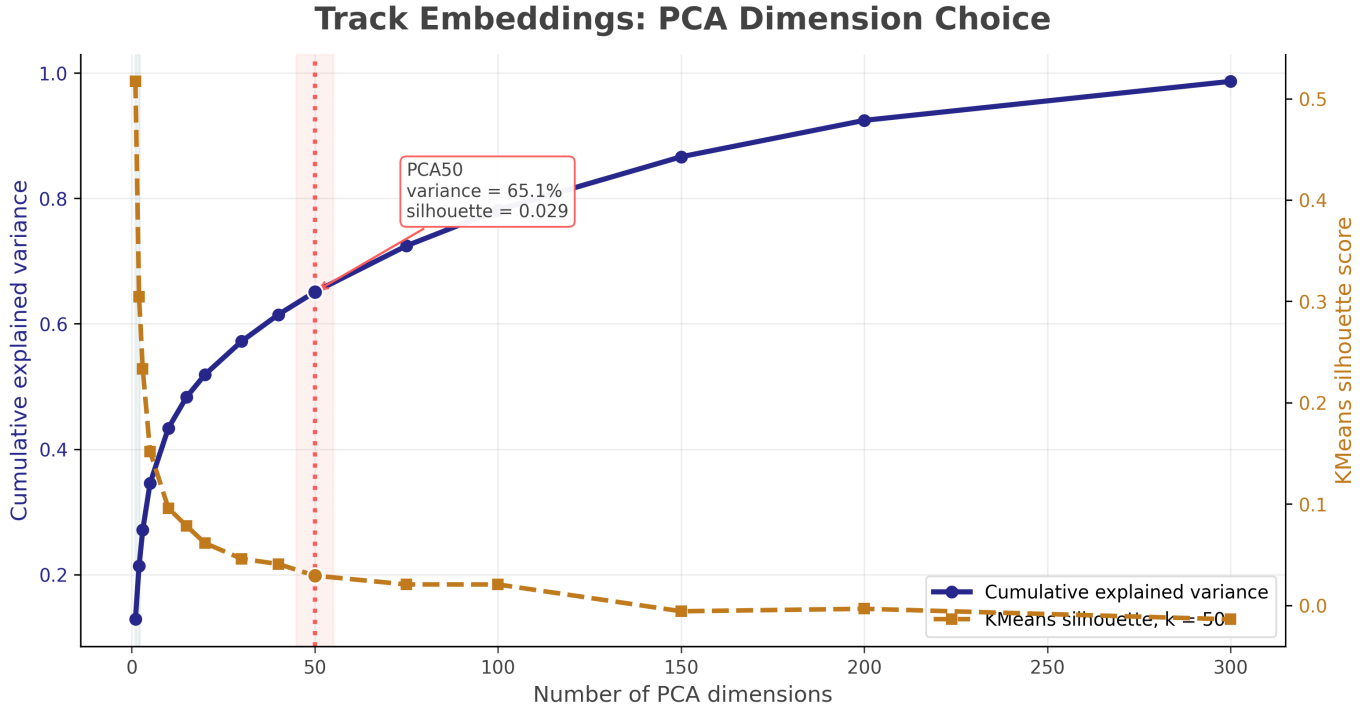


Fig. 6. Cumulative explained variance (related to the scree-profile behavior) and K-means silhouette score across PCA dimensions for the mean-pooled track embeddings. Very low-dimensional reductions yield higher apparent silhouette but retain little of the original variance. PCA-50 therefore provides a practical compromise, preserving about 65.1% of the variance while remaining suitable for downstream clustering and visualization.

and running playlists, country playlists, and christmas or holiday playlists. Because UMAP can distort distances and emphasize local structure, however, we used the 2D projection only for visualization and performed HDBSCAN clustering in the higher-dimensional UMAP-15d representation.

HDBSCAN tuning revealed a clear tradeoff between fragmentation and over-merging. The initial solution identified 104 clusters and labeled 12,580 playlists (62.9% of the 20,000-playlist sample) as noise, indicating that a conservative density threshold produced many small local clusters while leaving much of the dataset unlabeled. After tuning, the final solution identified 5 clusters and only 53 noise points (0.3%). However, this low-noise solution was highly imbalanced: one dominant cluster contained 19,480 playlists, while the remaining four clusters were much smaller, with sizes 315, 76, 50, and 26. Taken together, these two solutions suggest that the UMAP-based track embedding contains meaningful local density structure, but does not separate into many balanced, well-defined density clusters under HDBSCAN.

Figure 9 compares the initial and final HDBSCAN labelings on the same 2D UMAP display, making this tuning tradeoff visually explicit. In the initial solution, HDBSCAN fragments the embedding into many small local clusters while labeling much of the dataset as noise. In the final tuned solution, the noise rate drops almost completely, but most playlists are absorbed into one dominant cluster. HDBSCAN was therefore more useful here as a diagnostic of density structure than as evidence for a rich set of clearly separated playlist categories.

To assess whether the track-based groups remained interpretable, we examined playlist titles within clusters and

computed title coherence using average pairwise cosine similarity among title embeddings. The per-cluster title-coherence values varied substantially: the most coherent clusters had average title cosine similarities above 0.690, the highest was approximately 0.81, the least coherent clusters were near 0.220, and the overall mean was 0.368 with standard deviation 0.141. This supports a nuanced conclusion: some track-based groupings align strongly with playlist-title semantics, but the overall structure is uneven and the final HDBSCAN partition is dominated by one broad generic region rather than many balanced semantic clusters.

E. Similarity Retrieval

Finally, we evaluated whether the embedding space supported cold-start retrieval independently of dimensionality reduction and clustering. Using cosine similarity over normalized embeddings, we inspected nearest neighbors for several query playlists. Because human-titled playlists are inconsistently titled with symbols, spaces, and emojis, queried playlist titles are standardized. Queried titles have all extra white spaces removed and are lowercase, exactly as the embeddings in the large bank were. This allows for a stronger similarity between matching texts whereas white spaces, capital letters, and other symbols could distort the relationships. We also tried experimenting with two types of retrieval, title and songs.

The results were qualitatively plausible but should not be interpreted as a formal recommendation benchmark. Regarding title retrieval, if a user queries a playlist title such as "Throwbacks", then many similarly named playlists will be returned. This works well if there are a large number of

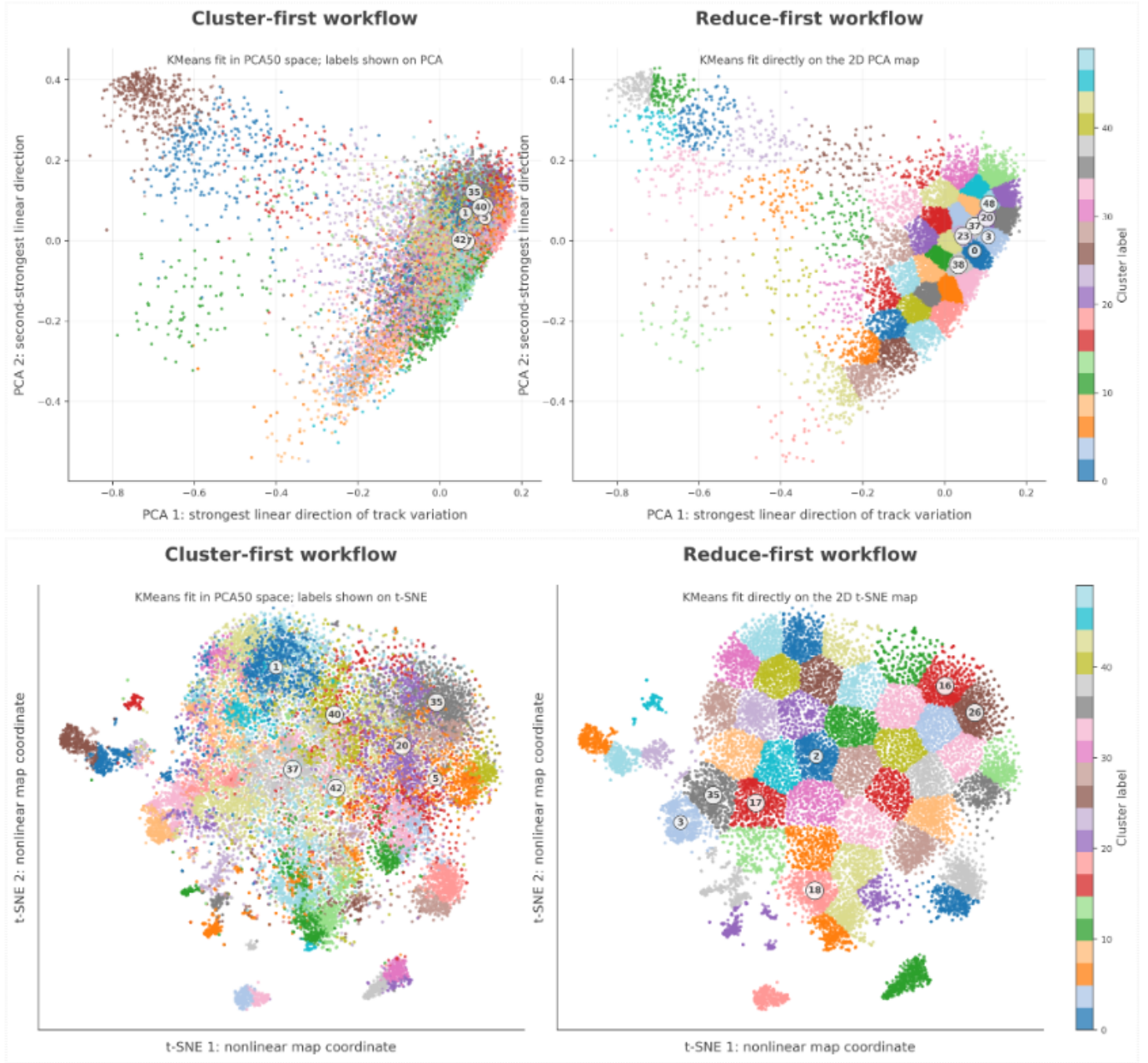


Fig. 7. Cluster-first versus reduce-first workflows on PCA and t-SNE embeddings. Left panels show K-means fit in PCA-50 and projected into 2D; right panels show K-means fit directly on the 2D PCA or t-SNE coordinates. Although the reduce-first workflow produces cleaner-looking clusters, it substantially changes the partition relative to the higher-dimensional reference.

playlists titled a specific word or phrase that people associate with an idea, such as “Throwbacks” relating to the early-2000s. However, if someone wanted to query a playlist such as “lively sunday evening on the beach” then the returned results always have beach in the title, not understanding the semantic meaning of the phrase. The cosine similarities for title retrieval are almost always above 0.8 if querying a common playlist title, meaning that this type of retrieval works very well for common playlist titles.

For track retrieval, querying a phrase works much better. Because the average embedding of a playlist has an overall semantic meaning that Sentence-BERT captures, we exploit it

to return more semantically meaningful songs in relation to the queried title. Querying the same “lively sunday evening on the beach” example returns songs more similar to the emotions expected from such a playlist title. In general, the cosine similarities are much lower for this method because exact phrases are muddled by averaging the playlist tracks. However, semantic meaning is preserved and allows for meaningful playlist retrieval.

These results suggest that tracks and titles have their own places in retrieval. Title retrieval emphasises the commonality of specific playlist titles while track retrieval excels at retrieving semantic meaning from a less common phrase or

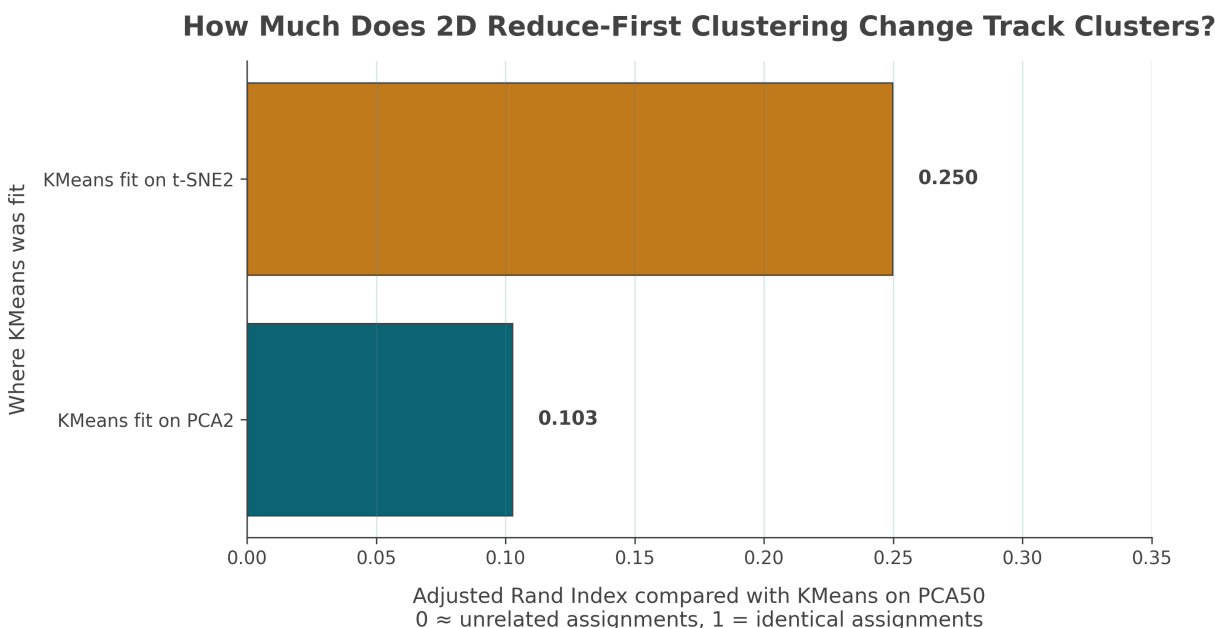


Fig. 8. Adjusted Rand index comparing reduce-first clustering on PCA-2 and t-SNE2 with the PCA-50 reference partition. Low ARI values indicate that clustering directly in two dimensions substantially changes the assignment structure.

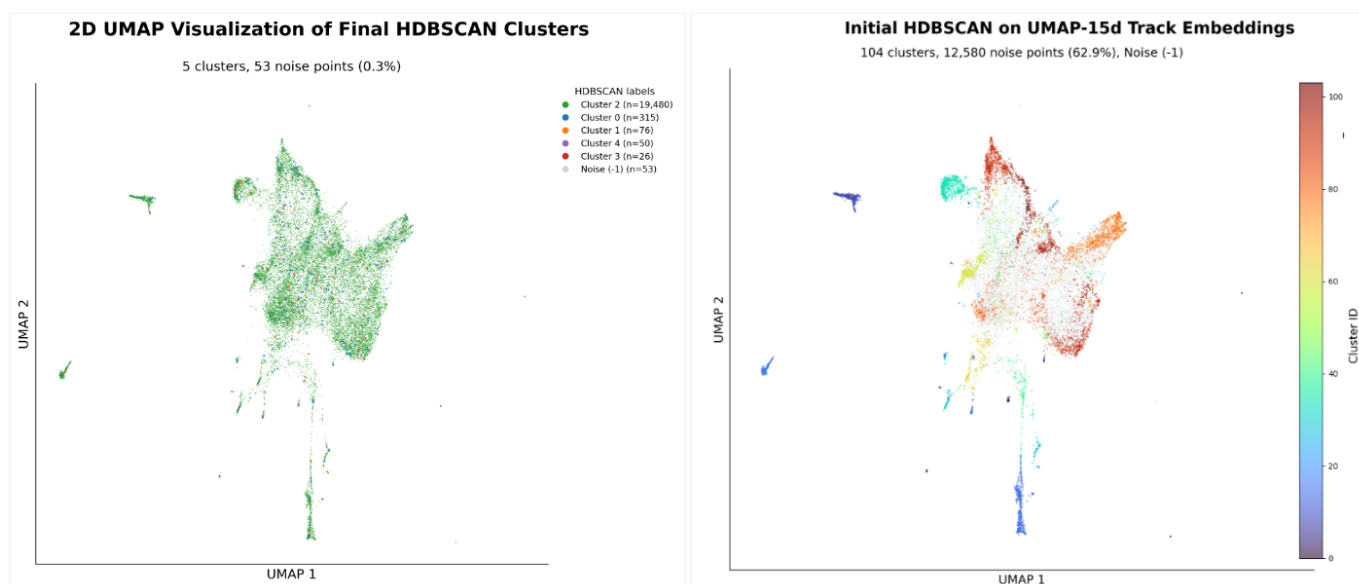


Fig. 9. HDBSCAN labels displayed on the 2D UMAP projection, with clustering performed in the higher-dimensional UMAP-15D space. The initial solution is highly fragmented, identifying 104 clusters and labeling 62.9% of playlists as noise, whereas the final tuned solution reduces noise to 0.3% and returns 5 clusters, largely by merging most playlists into one dominant group.

title. This means that the embeddings capture meaningful local semantic structure even without clustering. Accordingly, the main cold-start recommendation signal in this project comes from the geometry of the original embedding space, whereas dimensionality reduction and clustering are more useful for visualization and exploratory analysis than for nearest-neighbor retrieval itself.

However, the drawback with this method is the lack of music attributes when recommending. Since the embeddings are only of the titles of songs and playlists, any attributes of

the songs do not impact recommendation. Genre, beats per minute, artist, and many other qualities are all important to recommendation but are not used by this system. Furthermore, using the wrong retrieval type can return unusable results. For example, track querying "sad rock" returns only songs with the word "rock" in the title, disregarding the more important "sad" adjective to describe the songs. This system can be majorly improved by adding more information about songs to the embeddings.

VI. DISCUSSION

A. Semantic Signal in Playlist Titles

Our results suggest that playlist titles encode useful but incomplete semantic signal even though they are typically informal, short (median 7 characters), and written without descriptive intent. Exploratory statistics showed that many of the most frequent titles are concise genre, mood, or activity labels. This lexical simplicity, combined with SBERT’s extensive pretraining, explains why these terms can be embedded into a well-structured semantic space where related concepts like “workout,” “run,” and “gym” are geometrically proximal.

However, the main clustering representation was derived from mean-pooled track-title embeddings, so the observed structure reflects playlist content more directly than title wording alone. The alignment between title embeddings and track-based structure suggests that users are often consistent in their titling behavior: playlists called “chill” actually contain calmer, more ambient music than playlists called “workout.” This title-content alignment supports the key assumption behind title-based cold-start recommendation, but our results do not fully validate it as a complete recommendation system.

B. Comparison of Dimensionality Reduction Methods

PCA and UMAP capture complementary aspects of the embedding structure. PCA provides a globally interpretable linear projection that preserves variance but conflates clusters that differ in subtle, nonlinear ways. UMAP’s nonlinear projection reveals finer-grained structure—in particular, the smooth gradients between adjacent genres and the sharp separation of highly distinct ones—at the cost of distorting global distances. Neither method is strictly superior; the choice depends on the downstream task.

For recommendation, the raw 384-dimensional embedding space (with cosine similarity) outperforms any reduced representation because the reduction necessarily discards information. For this dataset, UMAP revealed more local density structure than PCA, but we treat that structure as exploratory and validate it using title coherence and manual cluster inspection. Methodologically, separating clustering from validation was essential: mean-pooled track embeddings were used to define the grouping structure, while title embeddings were used to assess whether those groups aligned with user-expressed semantic intent. This design avoids overinterpreting title-based structure while still testing the project’s cold-start motivation. More broadly, the PCA+UMAP pipeline combines a more stable reduced representation for clustering with a flexible nonlinear view for visualization.

C. Limitations

Several limitations constrain these findings. First, the embeddings were derived from track titles rather than audio features, so the system cannot distinguish songs that sound different but share similar text. Second, the dataset spans 2010–2017, and playlist naming conventions and genre patterns may have changed since then. Third, title embeddings and mean-pooled track embeddings have different geometric

properties: title embeddings lie close to a unit sphere, whereas mean-pooled track embeddings retain variable norms, making joint clustering difficult to interpret. In addition, title-based validation is indirect. Playlist titles can suggest semantic consistency within track-based clusters, but they do not provide formal genre or mood labels. Finally, the HDBSCAN results show that density-based clustering is sensitive to hyperparameter choice: conservative settings can produce high noise fractions, while more permissive settings can over-merge the data.

D. Future Research

Future work could extend this analysis in several directions. More systematic tuning of dimensionality reduction and clustering methods would help distinguish robust playlist structure from artifacts of a particular configuration, especially through comparisons of UMAP output dimensions, neighborhood sizes, distance metrics, and HDBSCAN hyperparameters. Future work could also improve the construction of track-based playlist embeddings. Mean pooling provides a simple and interpretable baseline, but alternative aggregation methods such as weighted pooling, attention-based pooling, or learned aggregation may better preserve distinctive songs and mixed-theme playlists [?]. Another useful direction would be to study how title-track alignment varies across playlist characteristics such as title length, playlist length, common title category, or cluster membership. Finally, a stronger recommendation system could combine text-based embeddings with audio features, artist metadata, genre labels, and user behavior, while using approximate nearest-neighbor methods such as FAISS or HNSW to scale retrieval to larger candidate sets.

VII. CONCLUSION

This project shows that mean-pooled SBERT track embeddings contain enough geometric structure to support exploratory dimension reduction and density-based clustering, while playlist-title embeddings provide a useful but incomplete validation and cold-start query signal. The track-embedding space groups some playlists in ways that align with human-interpretable title themes such as activity, mood, or genre-like labels, but this alignment varies across clusters. Nearest-neighbor retrieval over this space successfully recovers thematically related playlists using only a title as a query, providing a viable approach to cold-start recommendation.

From a mathematical visualization perspective, this project illustrates the complementary strengths of linear (PCA) and nonlinear (UMAP) dimensionality reduction on high-dimensional language model outputs, and shows how density-based clustering (HDBSCAN) can be used to diagnose local density structure in nonlinear projections. Together, these tools provide a useful exploratory pipeline from raw text embeddings to retrieval and visualization, even when the resulting density-based clusters are not cleanly balanced.

The final pipeline therefore does not claim that titles and tracks form one unified clustering space. Instead, it uses track content to define clusters and uses titles to test whether those clusters recover interpretable playlist intent.

APPENDIX A PROJECT REPOSITORY AND CODE FILES

The full project repository is available on GitHub at: <https://github.com/siddmohanty111/PlaylistClustering>

The repository contains the code, notebooks, and supporting files used to preprocess the Spotify Million Playlist Dataset, construct Sentence-BERT embeddings, tune clustering models, generate visualizations, and test the cold-start recommendation workflow. Because the raw Spotify dataset and generated embedding files are large, the repository is intended to document the computational workflow rather than store all intermediate data artifacts.

The main repository files used in this report are summarized below.

A. Embedding and Recommendation Code

`build_embeddings.py`

Creates the main embedding file used in the project. This script encodes playlist titles and track titles with Sentence-BERT and stores the resulting title embeddings and mean-pooled track embeddings in a row-aligned format.

`recc_utils.py`

Contains utility functions for the recommendation system, including cosine-similarity retrieval logic used to return playlists or songs from a title-based query.

B. Final Analysis and Visualization Notebook

`Math250_Spotify_Extended_Clustering_Analysis.ipynb`

Main visualization and analysis notebook for the final report. It contains code for comparing clustering workflows, generating cluster-quality figures and tables, plotting PCA and t-SNE results, and formatting UMAP/HDBSCAN visualizations.

C. Supporting Notebooks

`notebooks/hdbscan_tuning.ipynb`

Notebook used to tune HDBSCAN parameters and compare how different settings affect the number of clusters, noise fraction, and overall cluster structure.

`notebooks/testing_pipeline.ipynb`

Notebook documenting how the dataset and working files were moved into Google Drive and prepared for use in the shared project workflow.

`notebooks/title_to_tracks_pca.ipynb`

Early experimental notebook for the UMAP-HDBSCAN pipeline before final tuning. Although the filename refers to PCA, the notebook was used for exploratory dimensionality reduction and untuned UMAP/HDBSCAN clustering experiments.

D. Exploratory Analysis

`Math250ExploratoryData01.pdf`

Exploratory analysis output summarizing early dataset-level patterns, including playlist title lengths, playlist track counts, and other descriptive statistics used to motivate the paired title-track embedding design.

The notebooks in the repository should be interpreted as part of an iterative analysis process. Early exploratory notebooks document data access, preprocessing, and preliminary dimensionality-reduction experiments, while the final extended clustering notebook contains the visualizations and workflow comparisons used most directly in the report.

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